Disc Harrow Draft & Speed Calculations

Engineering Formulas and Field Metrics for Tractor HP Matching

1. ASABE STANDARD DRAFT FORMULA

ASABE D497.7 Standard Formula for Disc Harrows

Specific draft resistance depends on soil texture, working depth, and speed:

$$\text{Draft (N)} = F \times [\, F1 + (F2 \times \text{Speed}) + (F3 \times \text{Speed}^2) \,] \times \text{Width} \times \text{Depth}$$

Where: Width is in meters, Depth is in cm, Speed is in km/h, and F is a soil texture coefficient (Fine = 1.0, Medium = 0.70, Coarse = 0.45).

2. SPECIFIC DRAFT REFERENCE GUIDE

For rapid field estimations, use average force required per meter of working width at standard depth (7–10 cm / 3–4 inches):

Soil Type	Draft (kN / meter width)	Draft (lbf / foot width)	Recommended HP / Meter
Light / Sandy	3.0 – 4.5 kN/m	200 – 300 lbf/ft	12 – 18 HP / m
Medium / Loam	4.5 – 6.5 kN/m	300 – 450 lbf/ft	18 – 25 HP / m
Heavy / Clay	6.5 – 9.5 kN/m	450 – 650 lbf/ft	25 – 35 HP / m

3. POWER & SPEED CALCULATIONS

Drawbar Power to Engine HP Requirements

Convert calculated draft force and target working speed into required engine power:

$$\text{Drawbar HP} = [\, \text{Total Draft (lbf)} \times \text{Speed (mph)} \,] \,/\, 375$$

$$\text{Engine HP Required} = \text{Drawbar HP} \,/\, 0.67$$

4. PRACTICAL CALCULATION EXAMPLES

Example 1: Calculating HP required for a known disc size

- **Equipment:** 4.0-meter (13.1 ft) Tandem Disc Harrow
- **Field Conditions:** Medium Loam soil, 8 cm target depth, 8 km/h (5 mph) working speed
- **Step 1 (Draft):** 5.5 kN/m × 4.0 m = 22 kN (approx. 4,950 lbf total draft)
- **Step 2 (Drawbar Power):** (4,950 lbf × 5 mph) / 375 = 66 Drawbar HP
- **Step 3 (Engine HP Needed):** 66 / 0.67 = **98.5 Engine HP**

Example 2: Determining optimal speed for a 120 HP tractor

- **Tractor Available:** 120 Engine HP (~80 Drawbar HP available)
- **Equipment:** 5.0-meter (16.4 ft) Heavy Offset Disc in Clay Soil (7.5 kN/m draft = ~2,570 lbf/m)
- **Total Draft Load:** 2,570 lbf/m × 5.0 m = 12,850 lbf
- **Max Speed Formula:** Speed (mph) = (80 Drawbar HP × 375) / 12,850 lbf
- **Result:** Max speed = **2.33 mph (3.75 km/h)** — *Indicates implement is too wide for this tractor in heavy clay; reduce width or shallow depth.*

5. RULE-OF-THUMB SPEED & SIZING QUICK REFERENCE

Tractor Engine HP	Recommended Disc Width (Light Soil)	Recommended Disc Width (Heavy Clay)	Optimal Operating Speed
75 – 90 HP	3.0 – 3.5 m (10–11.5 ft)	2.0 – 2.5 m (6.5–8 ft)	7.5 – 9.5 km/h (4.5–6.0 mph)
100 – 130 HP	4.0 – 5.0 m (13–16.5 ft)	3.0 – 3.5 m (10–11.5 ft)	7.5 – 9.5 km/h (4.5–6.0 mph)
140 – 180 HP	5.5 – 7.0 m (18–23 ft)	4.0 – 4.5 m (13–15 ft)	8.0 – 10.0 km/h (5.0–6.2 mph)
200+ HP	7.5+ m (24.5+ ft)	5.0+ m (16.5+ ft)	8.5 – 10.5 km/h (5.3–6.5 mph)